01 · PROBLEM STATEMENT

Earth is 40 minutes away

On the ISS a flight surgeon is one radio call away. On Mars the delay is **4 to 40 minutes each way**, with blackouts NASA plans for of **up to two weeks** — and usually **no physician on board**.

> **Crews need an onboard system a non-medical crewmember can talk to in plain words – offline, space-aware, clear about when to call for help, and showing where every answer came from.**

What could this be? | **How serious is it?** | **What do we do now?**

02 · RESEARCH 1

There's no data to train on

NASA's own risk models say in-flight medical data is **severely limited** — for some serious conditions the in-flight count is **zero**. So VITALS grounds a pretrained model in **cited research** instead of training one.

CONDITION	WHAT SPACE CHANGES
Fluid-shift congestion	Day 1, looks like a cold — not an infection
Space motion sickness	Early flight only — day 60 means something else
Decompression sickness	Joint pain after an EVA outranks a strain
Renal stone	Bone loss + dehydration raise the risk
Jugular vein clot	Found on ISS with no symptoms — escalate red flags

35 conditions · 145 findings · every rule cited (NASA IMM, HRP evidence reports, flight-surgeon papers)

03 · CONSTRAINTS & REQUIREMENTS

ID	CONSTRAINT
C1	Fully **offline** — 40-min delay, 2-week blackouts
C2	User is **non-medical** and stressed
C3	Runs on a **Jetson Orin Nano 8 GB**
C4	Almost no in-flight medical data
C5	Health data is private
C6	Small screens, gloves, bad light

F1 plain words √ | F3 no invented conditions √ | F4 every source from the KB √ | F5 escalation 18/19 | F7 Wrong? button √ | F8 "nothing fits" — weakest | F10 PIN + server roles √ | F11 emergency button √

05 · RESEARCH 2 – MEASURED, NOT GUESSED

MODEL	RIGHT CONDITION	MISSED ESCALATIONS	TIME
llama3.2 3B	19/26 · 73%	**0**	~4 s
llama3.1 8B	22/26 · 85%	0	~8 s
llama3.2 1B	invented a symptom	backstop only	~80 s

33-prompt bench, 26 Sep 2026. The rules backstop took missed escalations from 3 → 0. Never demo on 1B.

TEAM TETHER

VITALS

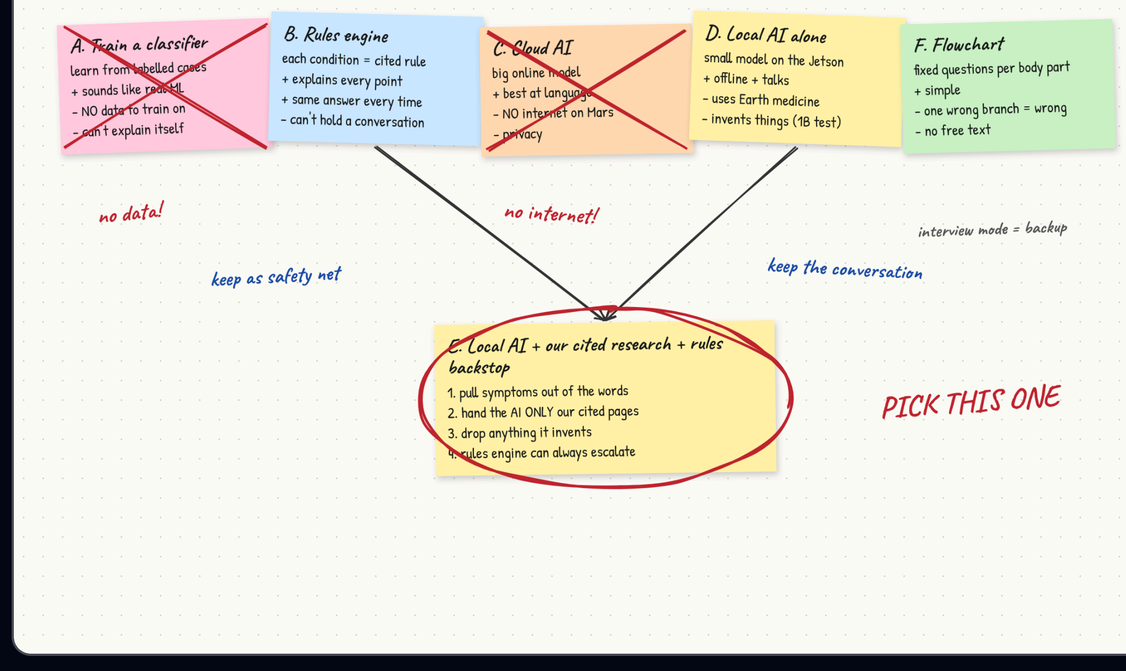
NASA project: Medical Diagnostic System with Machine Learning & Artificial Intelligence (From a non-Medical Person's Perspective)

South Texas ISD Science Academy · Mercedes, Texas · Team # _____

Joshua Collado · **Joaquin** · **Cruz** — Teacher: Ms. Sylvia Odoms

Scan for the website

04 · BRAINSTORMING 1 – THE BRAIN



Classifier × (no data) · Cloud AI × (no internet) · **Grounded local AI + rules ✓**

06 · BRAINSTORMING 2 – SCREENS & HARDWARE



Astronaut → ground crew → doctor · **Jetson server + 3 Pi screens ✓**

07 · FUNCTIONAL DIAGRAMS – THE THREE BEST

B · Rules only
Explainable, can't converse

MESSAGE → EXTRACT → SCORE → RANK

D · Local AI only
Talks well, invents things

MESSAGE → LOCAL LLM → FREE TEXT

E · Chosen ✓
Grounded AI + rules backstop

SCOPE (medical?) → EXTRACT (words → findings) → RETRIEVE (cited KB pages) → REASON (local AI reads only those) → CHECK (drop anything invented) → ANSWER (level · why · what to do · sources)

KNOWLEDGE BASE (35 cited conditions)

ENGINE – RULES BACKSTOP (escalates if the model OR the engine says so)

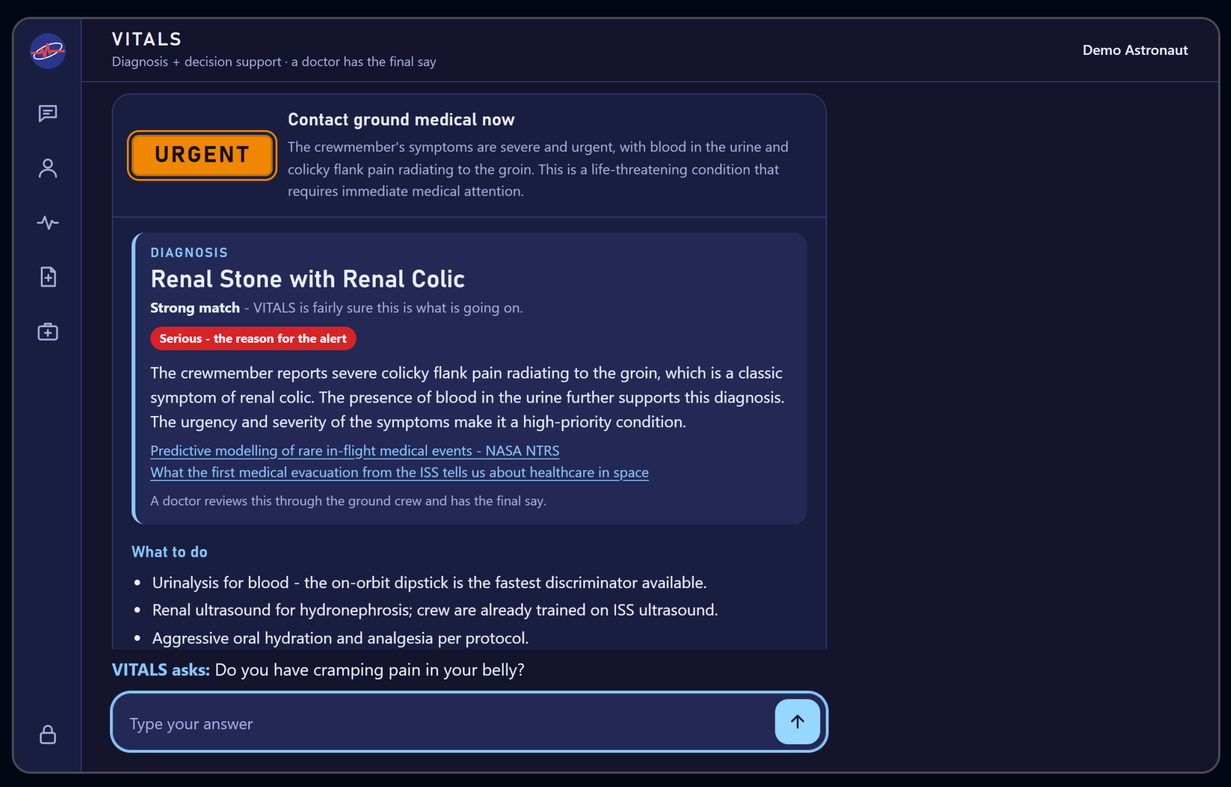
MISSION LOG (every step · subsystem · status · time)

08 · DECISION MATRIX

CRITERION	WT	A	B	D	E
Fully offline	5	5	5	5	**5**
Never misses an escalation	5	2	4	2	**5**
Shows its sources	4	1	5	2	**5**
Understands real speech	4	3	2	5	**4**
Works with the data that exists	4	1	4	4	**4**
Runs on the Jetson	3	5	5	3	**3**
Testable, repeatable	3	3	5	2	**5**
Buildable this year	2	2	4	5	**3**
Total / 150		83	127	104	**132**

A classifier · B rules · D local AI alone · E grounded AI + rules. Model: llama3.2 3B (69/85). Deployment: Jetson + 3 Pis (wins).

09 · FIGMA + CODE – THE ANSWER BOX




```
# src/vitals/server.py – run_pipeline() (trimmed)
scope = classify(latest, knowledge_base)        # SCOPE
observations = KeywordExtractor().extract(      # EXTRACT
    complaint, knowledge_base)
retrieved = retrieve(knowledge_base, complaint, # RETRIEVE
    observations, limit=RETRIEVAL_LIMIT)
answer = ask(knowledge_base, complaint,         # REASON + CHECK
    retrieved, observations=observations,       # + ENGINE backstop
    model=model)
```

Figma by Cruz is the idea; the built screen follows it. Same steps run in **vitals ask**, so the screen can't drift from what's tested.

12 · STORYBOARD — "IT COMES IN WAVES"

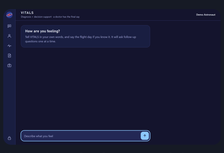
Commander Rivera, flight day 63 on a Mars transit. Ground is 18 minutes away.

1



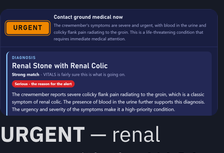
Enters her PIN

2



"Bad pain in my side, comes in waves… blood when I pee"

3



URGENT — renal stone, with the NASA source

4

Taps **Wrong?** on something it misread — the answer updates

5

One follow-up question at a time, never repeated

6 · PLANNED

Ground crew redacts the case, sends it to the doctor, who agrees and adds a note

11 · PRELIMINARY TEST PLAN

LEVEL	PASS WHEN	NOW
Unit — every module	all pass	262 ✓
Missed escalations	must be 0	1 of 19
Vignettes — 60 real-speech prompts	hits ≥ 80%	73–85%
Offline — network unplugged	answers, page loads	✓
User testing — 5+ people	level read in ≤ 3 s	after PDR
Jetson timing	≤ 15 s an answer	Nov

16-17 · MISSIONS & LOGS

6/6
BASIC MISSIONS

18/19
MUST-ESCALATE (A2)

0.3 s
BASIC RUN TIME

```
2026-10-17T22:33:50Z  KB         kb/           PASS
2026-10-17T22:33:50Z  SENSORS    sensors.py    NOT_CONNECTED
2026-10-17T22:33:50Z  SCOPE      "hi"          PASS
2026-10-17T22:33:50Z  ENGINE     29 cases      PASS
2026-10-17T22:33:50Z  CASES      owner check   PASS
2026-10-17T22:33:50Z  MISSION    run           OK  B1–B6
```

Timestamp · subsystem · element · status — one line per step. The miss: pain-only bends after an EVA. Fix in progress.

60% PROTOTYPE · NEXT STEPS

~62% built — weighted by subsystem

Nov	Flash the Jetson, time an answer · fix "nothing fits"
Dec–Feb	Ground crew + doctor screens · first sensor · user testing
Mar · CDR	90% prototype · voice input
May · FDR	Johnson Space Center

VITALS is a student prototype for decision support — a doctor has the final say. Not a medical device.